

ON A MODEL FOR SPARES SUFFICIENCY

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Key words: Number of repairs; optimal strategy; probability of spares sufficiency.

Summary: A new method of assessing the probability of spares sufficiency is developed for the components that can be repaired a finite number of times. An optimal strategy is introduced for the general case according to which “the best” available components of a system should be in the operating state of every instant of time. An approximate approach for applications is considered and a simple example is presented.

1. Introduction

Consider a component with a continuous distribution function (DF) of time to failure $F(t)$ and n spares identical to the initial component. It starts functioning at $t = 0$ and is replaced upon failure by a spare, etc. The process ends when the operating component fails and no spares are left. The time of replacement is assumed to be negligible and spares are supposed not to age while waiting for operation. Under these assumptions the DF of time to failure

of the system (component + n spares) is defined in the usual way as the n -fold convolution of $F(x)$:

$$1 - \gamma_n(t) = F * F * \dots * F(t) \quad (1)$$

where the complementary probability $\gamma_n(t)$ is called the probability of spares sufficiency in $(0, t]$.

For the exponential DF: $F(t) = 1 - e^{-\lambda t}$ a well-known (in applications) formula exists (see, for instance, Ushakov, 1994):

$$\gamma_n(t) = e^{-\lambda t} \sum_{i=0}^n \frac{(\lambda t)^i}{i!}. \quad (2)$$

Assume now that spares are intended for $k \geq 1$ identical functioning components in series. It can be immediately seen from the properties of the exponential DF that equation (2) becomes

$$\gamma_n^{(k)}(t) = e^{-k\lambda t} \sum_{i=0}^n \frac{(k\lambda t)^i}{i!} \quad (3)$$

while in general, a reasonably simple expression cannot be obtained. For the case of two components in series and one spare, the following simple formula can still be derived in a straightforward way:

$$\begin{aligned} \gamma_1^{(2)}(t) &= (1-F(t))^2 + 2 \int_0^t f(x)(1-F(x))(1-F(t-x))(1-F(t-x|x))dx \\ &= (1-F(t))^2 + 2(1-F(t)) \int_0^t f(x)(1-F(t-x))dx \end{aligned} \quad (4)$$

where $f(x) = F'(x)$ and $F(t|x)$ denotes the component's remaining lifetime DF (on condition that it had survived in $(0, x]$):

$$F(t | x) = \frac{F(t+x) - F(x)}{1 - F(x)}.$$

For DFs other than the exponential, deriving $\gamma_n^{(k)}(t)$ for $k > 2$, $n \geq 2$ can result in cumbersome formulas requiring the use of numerical methods to obtain results in practice.

The described model deals with the situation when the failed component is replaced by a spare one and is not repaired afterwards. Assume now that each component can be repaired $r \geq 0$ times before "the final failure". This often takes place in maintenance of modern electronic equipment (chips for computer processors, for instance). Some of the failures are easily and quickly repaired and the item can afterwards be put into operation again immediately or considered as a spare one. Assume also that after each repair the component is characterized by the same time to failure DF $F(t)$. The described practical pattern leads to a problem of assessing the probability of spares sufficiency, which was not addressed in the literature before. It is clear that the proper model for considering these additional repairs will substantially decrease the number of spares (and corresponding costs), especially for long periods of time.

2. The Model

As in the Introduction, consider a system of $n+1$ identical components (having $F(t)$ as time to failure DF): one initial and n spares. A failed component can be repaired (instantly) with probability $1 - p_s$ and is unrepaired (final failure) with complementary probability p_s , where s is the number of the failure, $1 \leq s \leq r$. Thus r is the maximum number of repairs that can be performed on each of the $n+1$ components. Technically this setting means that certain

faults cannot be repaired while others can, but the number of repairs of a given component is bounded.

As in Finkelstein (1991) it is reasonable to assume that $\{p_s\}$ are non-decreasing in s :

$$p_1 \leq p_2 \leq \dots \leq p_r < p_{r+1} = 1. \quad (5)$$

Indeed, very often in practice the probability that the occurred failure of the component is the final one, increases with the number of this failure.

The following theorem is rather straightforward.

THEOREM 1. The probability of spares sufficiency in $(0, t]$ can be assessed using formula (1), where the governing DF $F(t)$ is replaced by the equivalent one, as given in (7) below.

PROOF: The real procedure of a system's functioning can be rather complicated: after the first failure (that can be repaired) the component is replaced by a spare one, the failed one is repaired and starts to be a spare itself etc. But it is easily seen from a probabilistic point of view that in terms of assessing the probability of spares sufficiency the real scenario can be converted into the following: Initial component starts functioning at $t = 0$ and is replaced only after the final failure (when no repairs can be performed), then the first spare starts functioning etc. The system fails, when the last component fails and the DF of time to this final event does not depend on the sequence in which components were functioning, given that the repair time is assumed to be negligible. Thus, the generalization of equation (1) is as follows:

$$\begin{aligned}
\gamma_{n,r}(t) &= (1 - F_r(t)) + \int_0^t f_r(x)(1 - F_r(t-x))dx \\
&+ \int_0^t f_r^{(2)}(x)(1 - F_r(t-x))dx + \cdots + \int_0^t f_r^{(n)}(x)(1 - F_r(t-x))dx
\end{aligned} \tag{6}$$

where

$$\begin{aligned}
f_r^{(i)}(x) &= \int_0^x f_r^{(i-1)}(z)f_r(x-z)dz, \quad i = 2, \dots, n; \\
f_r^{(1)}(x) &\equiv f_r(x) ; \quad f_r(x) = F_r'(x)
\end{aligned}$$

and $F_r(t)$ is defined in a straightforward way through :

$$\begin{aligned}
1 - F_r(t) &= (1 - F(t)) + \int_0^t f(x)(1 - p_1)(1 - F(t-x))dx \\
&+ \int_0^t f^{(2)}(x)(1 - p_1)(1 - p_2)(1 - F(t-x))dx + \cdots \\
&+ \int_0^t f^{(r)}(x) \prod_{s=1}^r (1 - p_s)(1 - F(t-x))dx
\end{aligned} \tag{7}$$

where

$$\begin{aligned}
f^{(i)}(x) &= \int_0^x f^{(i-1)}(z)f(x-z)dz, \quad i = 2, \dots, r ; \\
f^{(1)}(x) &\equiv f(x) .
\end{aligned}$$

Thus equation (7) shows "the structure" of the equivalent DF $F_r(t)$ which can be substituted into equation (6) to give an expression for the probability of spares sufficiency.

In practice the spares sufficiency is analyzed and spares assessment (including optimal) is usually performed for the exponential DFs. In this case:

$$1 - F_r(t) = e^{-\lambda t} + e^{-\lambda t} \lambda t (1 - p_1) + \dots + e^{-\lambda t} \frac{(\lambda t)^r}{r!} \prod_{s=1}^r (1 - p_s). \quad (8)$$

Specifically, when $p_s = p$, $s = 1, 2, \dots, r$:

$$\begin{aligned} 1 - F_r(t) &= e^{-\lambda t} \sum_{i=0}^r \frac{(\lambda t (1-p))^i}{i!} \\ &= e^{-\lambda t p} \left(e^{-\lambda t (1-p)} \sum_{i=0}^r \frac{(\lambda t (1-p))^i}{i!} \right). \end{aligned} \quad (9)$$

If $p = 0$, which means that each failure up to the r th one can be repaired, then equation (9) simplifies to

$$1 - F_r(t) = e^{-\lambda t} \sum_{i=0}^r \frac{(\lambda t)^i}{i!}. \quad (10)$$

It can be easily seen that equation (6) then becomes

$$\gamma_{n,r}(t) = e^{-\lambda t} \sum_{i=0}^{(n+1)r-1} \frac{(\lambda t)^i}{i!}. \quad (11)$$

Expression (11) can be interpreted as the probability of sufficiency for the component with failure rate λ having $(n+1)r - 1$ identical spares.

One cannot obtain the same simple results for $p_s = p \neq 0$ but, as follows from equations (9) and (6), the eventual formula for $\gamma_{n,r}(t)$ in this case can be derived in a straightforward way. It is worth mentioning that the helpful and simple equation for the mean value of the random variable, described by the general DF (7) is:

$$T_r = \int_0^{\infty} (1 - F_r(t))dt = p_1 T + p_2(1 - p_1)(2T) + \dots \quad (12)$$

$$+ p_r \left(\prod_{s=1}^{r-1} (1 - p_s) \right) rT + \left(\prod_{s=1}^r (1 - p_s) \right) (r+1)T$$

where T is the mean time to failure for the initial DF: $T = \int_0^{\infty} (1 - F(x))dx$.

Indeed, one need not integrate equation (7) to conclude that the first term in the right-hand part of (12) is the probability that the first failure is the final one times the corresponding mean duration, the second term is the probability that the second failure is the final one times the corresponding mean duration (two cycles) etc.

As in the Introduction, let the spares be supplied now for $k > 1$ identical components, functioning in series and each component, including spares, can be repaired according to the same pattern (5) similar to the case of $k = 1$. As it was mentioned and as it follows from the simplest example (4), the exact formulas start to be very complicated for the initial components functioning in series. That is why approximate approaches can be very helpful in dealing with these problems and we shall refer to them in the next section. Along with it the

strategy of a system's functioning for $k > 1$ starts to be very important and we shall now mainly focus on its discussion. When $k = 1$ the sequence of components functioning is not important for assessing $\gamma_{n,r}(t)$ under given assumptions. That is why the simplest strategy (in terms of probabilistic description) was chosen: the initial component is replaced by a spare one only after its final failure etc. For $k > 1$ this sequence is important and the key for the best strategy will be given by Theorem 2.

But at first we shall look at a simple supplementary result. Consider a system of two identical components (main and standby) of the type described above. The first component starts operating at $t = 0$, while the second one is in a cold standby state. After a total failure of the first component, the second starts operating immediately. The switching from the operating component to a standby one can be instantaneously performed at any moment of time. The problem is to find the switching strategy that under given assumptions maximizes probability of a system's performance without total failures (of components) in the given time interval. It is equivalent to maximizing the random time to the first total failure which should be understood in terms of stochastic ordering (see later). The following is quite obvious.

LEMMA: An optimal strategy for the described system is the one which places the component with the maximal number of repairs left at the operating position.

PROOF: This strategy is realized as follows: the first component starts functioning and is replaced by a standby one after the first failure. The failed component is repaired and is aimed to replace the functioning one after its failure etc. It can be easily seen that if

$$p_s = 0, \quad s = 1, 2, \dots, r; \quad p_{r+1} = 1 \quad (13)$$

then any strategy that leaves the remaining component in the state with no repairs left after the total failure of the other component will be optimal. Indeed, the number of i.i.d. cycles to the first total failure of the component is maximal in this case and equal to $2r + 1$. The simplest strategy of this kind is the following: the first component starts operating and is replaced by the standby only after the r -th failure; the former standby operates till its total failure. It is clear that the strategy, defined by the Lemma, is also optimal.

For the general case (5) only "one by one" (in turn) strategy of the Lemma will be optimal. It can be clearly seen, considering the following expression for the probability of a system's functioning without total failures of components in $(0, t]$, $\forall t$ under some arbitrary strategy:

$$\begin{aligned} P_{arb}(t) = & \tilde{p}_0(0, t) + \tilde{p}_1(0, t)(1 - p_1) \\ & + \tilde{p}_2(0, t)(1 - p_1)(1 - p_{j_1}) + \tilde{p}_3(0, t)(1 - p_1)(1 - p_{j_1})(1 - p_{j_2}) \\ & + \dots + \tilde{p}_{2r}(0, t) \left(\prod_{i=1}^r (1 - p_i) \right)^2 \end{aligned} \quad (14)$$

where $\tilde{p}_i(0, t)$ is the probability of i events $0 \leq i \leq 2r$ in the renewal process with $F(t)$ as a governing DF and j_k , $k = 1, 2, \dots, r$ are integers: $1 \leq j_k \leq r$ with obvious restrictions on their values. It is clear that due to (5), every term in (14) from the third till the $2r$ -th is maximized by the one by one strategy. For example:

$$\max_{j_1} (1 - p_{j_1}) = 1 - p_1$$

$$\max_{j_1, j_2} (1 - p_{j_1})(1 - p_{j_2}) = (1 - p_1)(1 - p_2)$$

and this completes the proof.

THEOREM 2: Let $F(x) = 1 - e^{-\lambda x}$. Then the best (optimal) strategy for $k > 1$ is the one when the best available spare component is chosen for replacement each time the failure of a functioning component occurs.

PROOF: The optimal strategy is defined here as the strategy that maximizes $\gamma_{n,r}^{(k)}(t)$. The best component is defined as having the maximal number of possible repairs left. The components are considered to be equivalent if they have an equal number of possible repairs left. In this case an arbitrary component can be chosen for replacement. Thus all $k + n$ components of the system can be ordered in the process of functioning.

We shall prove this theorem for $k = 2$, $n = 1$ when assumption (13) holds. The generalization on assumption (5) is easily performed using considerations of the Lemma while the case $k \geq 2$, $n > 1$ can be treated in a similar way. Thus the system of two components operating in series and one standby is under investigation. Let T_f denote a random time to failure of the system. Our goal is to find the strategy S_{opt} which maximizes $\gamma_{1,r}^{(2)}(t)$ for each t . From the definition of stochastic ordering (indicated as $\leq_{st}$) in Hoyland and Rausand (1994) it follows that in this case :

$$T_{f,arb} \leq_{st} T_{f,opt}$$

where $T_{f,arb}$ and $T_{f,opt}$ denote time to system failure under the arbitrary and optimal strategy respectively.

Consider some arbitrary realization when, for instance, the third component had experienced the total failure first, thus leaving the first and the second components functioning until the total failure of either of them (failure of the system).

It can be more convenient at a certain stage to describe the situation equivalently in terms of random numbers of i.i.d cycles having $1 - e^{-2\lambda x}$ as the DF of the cycles' duration (each cycle ends with a failure of some component). Let the third component have totally failed at the end of the K_3 -th cycle of the system's functioning. Thus

$$K_3 = (r + 1) + r_1 + r_2$$

where $r_1(r_2)$ denotes the number of cycles when the first (second) component had failed: $0 \leq r_1, r_2 \leq r$. This means that $r_1 + r_2$ is fixed for the realization under consideration. Now we shall introduce the strategy for the first K_3 cycles that will maximize the system's remaining lifetime after the failure of the third component. Only $r_1 + r_2$ cycles where there were failures of the first and the second components will be controlled. Our goal is to arrive at the end of the K_3 -th cycle with the minimal difference between $r - r_1$ and $r - r_2$: It will be shown at the end of the proof that

$$\min |r_1 - r_2| \Rightarrow \max_{r_1 + r_2 = c, \forall t} \sum_{i=1}^{r-r_1} e^{-\lambda t} \frac{(\lambda t)^i}{i!} \sum_{i=1}^{r-r_2} e^{-\lambda t} \frac{(\lambda t)^i}{i!} \quad (15)$$

where c is constant which means that the remaining lifetime is maximized. What kind of strategy, applied to the first and the second component (the realization for the third component is fixed), can achieve $\min |r_1 - r_2|$? It is obvious that if we place these components in the "in turn" pattern on the $r_1 + r_2$ controllable cycles this goal will be achieved as this difference can be either 0

or 1. We had arbitrarily assumed for realization that the third component had failed first, but it can be the other one. Hence, S_{opt} should be the one that performs the in turn rule for each pair of components (considered separately apart from the third one). Taking into account the symmetry in this setting, we arrive at the conclusion that the strategy defined by this theorem is optimal.

Coming back to the proof of (15), it is clear that it will be valid if $\hat{p}_m \hat{p}_e$ is increasing as the difference between m and ℓ ($m + \ell = c$) is decreasing, where $\hat{p}_m$ ($\hat{p}_e$) denote the following sum:

$$\hat{p}_m = \sum_{i=0}^m \frac{(\lambda t)^i}{i!}.$$

Let $\Delta = \hat{p}_{m+1}\hat{p}_{\ell-1} - \hat{p}_m\hat{p}_\ell$. Then

$$\begin{aligned} \Delta &= \left(\hat{p}_m + \frac{(\lambda t)^{m+1}}{(m+1)!} \right) \left(\hat{p}_e - \frac{(\lambda t)^\ell}{\ell!} \right) - \hat{p}_m \hat{p}_e \\ &= \frac{(\lambda t)^{m+1}}{(m+1)!} \left(\frac{(\lambda t)^{\ell-1}}{(\ell-1)!} + \dots + \lambda t + 1 \right) - \frac{(\lambda t)^\ell}{\ell!} \left(\frac{(\lambda t)^m}{m!} + \dots + \lambda t + 1 \right). \end{aligned} \quad (16)$$

It is sufficient to prove that if $(m+1) < \ell$, then $\Delta > 0$ (if $m+1 = \ell$, then $\Delta = 0$, and if $m+1 > \ell$, then $\Delta < 0$).

From (16):

$$\begin{aligned} \Delta &= (\lambda t)^{m+\ell} \left(\frac{1}{(m+1)!(\ell-1)!} - \frac{1}{m!\ell!} \right) + \\ &\quad \sum_{k=1}^m (\lambda t)^{m+\ell-k} \left(\frac{1}{(m+1)!(\ell-1-k)!} - \frac{1}{(m-k)!\ell!} \right) + \\ &\quad \sum_{k=m+1}^{\ell} \frac{(\lambda t)^{m+\ell-k}}{(m+1)!(\ell-1-k)!}. \end{aligned} \quad (17)$$

Due to $(m+1) < \ell$, the first term in the right-hand side of (17) is positive. It is also obvious that for $k = 1, 2, \dots, m$:

$$\frac{(m-k)!\ell!}{(m+1)!(\ell-1-k)!} > 1$$

which completes the proof of the theorem.

One can interpret the result of Theorem 2 in the following way: the optimal strategy uses the best components as early as it can. Thus when termination of the system's life of the described type occurs, the remaining components are stochastically more worn-out than under any other strategy. When $k = 1$ the system fails when the last component fails and there are no remaining components left. The assumption that $F(x)$ is exponential was important for this proof for obtaining identical DFs of the times between steps.

For considering the general case of arbitrary $F(x)$, an assumption of the increasing failure rate of this DF should be made. This, along with (5), leads to some ageing properties that can form the basis of the future proof.

3. Practical application

In practical applications the assumption of constant failure rate is usually made. The exponential distribution is the most commonly used lifetime distribution in applied reliability analyses. The reason for this is its mathematical simplicity and that it leads to realistic lifetime models for certain types of units. According to Hoyland and Rausand (1994), various electronic systems are good examples of equipment with constant failure rate.

Equation (2) is the basic one for methods of optimal sparing with (or without) restrictions for systems with different types of components. One can use expression (11), obtained for the specific case when $p_s = p = 0$, $s = 1, 2, \dots, r$, $p_{r+1} = 1$, for applying these methods as well, but for the

general case even under exponential assumptions (8), the procedure becomes very complicated. That is why simple approximate approaches can be very helpful. The simplest one is based on equation (12) for the mean T_r of the equivalent DF, $F_r(t)$: each component with the equivalent DF is substituted by a component with exponential DF, having the same mean:

$$\tilde{F}_r(t) = 1 - e^{-\frac{t}{T_r}} = 1 - e^{-\lambda_r t} \quad (18)$$

where $\lambda_r = 1/T_r$ is the corresponding failure rate.

Then $\gamma_{n,r}(t)$ can be derived in the ordinary manner similar to (2):

$$\gamma_{n,r}(t) \approx e^{-\lambda_r t} \sum_{i=0}^n \frac{(\lambda_r t)^i}{i!} \quad (19)$$

The reverse problem can be easily solved, meeting the specified probability of spares sufficiency for one type of components and the case of different types of components (optimal spares) can be also approached using the approximate equality (19). In this paper we considered, for simplicity, only the case of components of one type. The error in (19) will be investigated in future work but some useful bounds can be already used, taking into consideration that DF (8) belongs to the increasing failure rate type (as follows from Barlow and Proschan, 1975) due to (5) and the fact that Erlangian DF has an increasing failure rate.

Approximation (19) can be also applied in practice for $k > 1$ resulting in a formula similar to (3):

$$\gamma_{n,r}^{(k)}(t) \approx e^{-k\lambda_r t} \sum_{i=0}^n \frac{(k\lambda_r t)^i}{i!} \quad (20)$$

Formula (20) is a very rough one. At the moment the authors do not know how to estimate its error. It "does not feel" the impact of the optimal

strategy on $\gamma_{n,r}^{(k)}(t)$. But it can be used for obtaining comparative results, showing "reliability growth" resulting from the possibility of repairs of the failed component.

On the other hand, by applying the optimal strategy we obtain the maximum of $\gamma_{n,r}^{(k)}(t)$ without knowing its exact value in this case. It is still worthwhile doing so because increasing a system's reliability is a very important task in practical applications.

EXAMPLE: Consider the real practical example of spares assessment for the specific electronic equipment that is to function for a period of 25 000 hours.

$$\begin{aligned} \text{Let } k &= 15, \quad r = 3; \\ p_1 &= 0,1, \quad p_2 = 0,3, \quad p_3 = 0,5 \\ \lambda &= 40 \cdot 10^{-6} 1/hr \implies T = 25000hr \end{aligned}$$

Then from equation (12):

$$\begin{aligned} T_3 &= 0,1 \cdot 25000 + 0,3 \cdot 0,9 \cdot 50000 + 0,9 \cdot 0,7 \cdot 0,5 \cdot 75000 \\ &\quad + 0,9 \cdot 0,7 \cdot 0,5 \cdot 10000 \approx 72000hr. \end{aligned}$$

Solving the elementary reverse problem for 3 different given values of spares sufficiency:

$$\begin{aligned} \gamma_{n,3}^{(15)}(25000) &= 0,9 \implies n = 7 \\ \gamma_{n,3}^{(15)}(25000) &= 0,99 \implies n = 10 \\ \gamma_{n,3}^{(15)}(25000) &= 0,999 \implies n = 12 \end{aligned}$$

The corresponding numbers of spares without repairs are 18, 22, 26 respectively. Thus the decrease in the needed numbers of spares due to the possibility of repairs is clearly seen.

REMARK:

A very useful index can be introduced based on the definition (12):

$$R = \frac{T_r}{T} = p_1 + 2p_2(1 - p_1) + \cdots + r p_r \prod_{s=1}^{r-1} (1 - p_s) \\ + (r + 1) \prod_{s=1}^r (1 - p_s),$$

showing the increase in the mean time to failure of the component due to the possibility of repairs. For the given example $R \approx 2,8$.

4. Conclusion

The main idea of the method presented in this paper, is to introduce the equivalent DF, $F_r(t)$ for the component that can be repaired r times. It gives accurate results for assessing the probability of spares sufficiency when this assessment is made for one component and approximate results for $k > 1$ components in series. An optimal strategy for $k > 1$ is defined as the strategy that places the components with maximal available repairs left in the operating state. It seems that this is the specific case of the general reliability rule in the presence of “internal” termination: the system had failed but some components are still operable and the goal of the strategy is to achieve the maximum of the wear-out of these remaining components.

The model leads to a substantial reduction in the numbers of needed spares, compared with an assessment that is not based on the possibility of r repairs.

Acknowledgements

The authors want to thank an anonymous referee and the editor whose comments had greatly improved the presentation of this paper and also Verónica Esaulova, from St. Petersburg University, for the short proof of relation (14).

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